MIL-E-1/973B
2 December 1970
SUPERSEDING
MIL-E-1/973A
23 February 1965

MILITARY SPECIFICATION SHEET

ELECTRON TUBE, RADIATION COUNTER

TYPE 5980

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

DESCRIPTION: Halogen-filled, low sensitivity, self-quenching for detection of gamma radiation

Dimensions:	See figure 1
Mounting position:	Any
Weight:	.22 ounce, nominal

ABSOLUTE RATINGS:

Parameter:	Ebb	TA	Rp
Unit:	Vdc	°C	Meg
Maximum:	725	75	---
Minimum:	675	-40	---
TEST CONDITIONS:	700	---	1.0

GENERAL:

Qualification - Required (See note 2)

Marking - See note 3

Response characteristics - See note 1

(B) denotes changes

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METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS		UNIT
						MIN	MAX	
	Quality conformance inspection, part 1							
6201	Background, contamination and photosensitivity	t = 2 minutes (see notes 5, 6, and 10)	0.65	II	N/t	---	10	Npm
6211	Starting voltage	Pulse amplitude = 1 volt; Nps = 200 (max); R2 = 1 Meg; R1 = C1 = C2 = 0; C3 = 0.01 μF (see note 10)	0.65	II	Es	---	640	Vdc
6216	Relative plateau slope	Voltage range = 660 to 760 Vdc (see note 12)	0.65	II	Ps	---	0.2	%/Vdc
6216	End-of-plateau voltage	Ps = 0.3 %/Vdc (see note 12)	0.65	II	Ee	780	---	Vdc
6221	Response count rate and current (gamma)	t = 2 minutes (see notes 7 and 10)	0.65	II	N/t	75	125	Nps
	Quality conformance inspection, part 2							
6221	Response count rate and current	Gamma (radium); rate = 500 mr/hr; Ebb = 710 Vdc (see note 10)	---	---	Ib	---	10.0	μAdc
6226	Pulse amplitude (1)	Ebb = 675 Vdc; Nps = 200 (max); R1 = 0.9 Meg; R2 = 0.1 Meg; C1 = 330 pF; C2 = 3,000 pF; C3 = 0.01 μF; multiply oscilloscope reading by 10 (see note 10)	---	---	eo	2.5	---	v
6226	Pulse amplitude (2)	Ebb = 725 Vdc; pulse amplitude (1)	---	---	eo	---	65	v
	Quality conformance inspection, part 3							
---	Life-test provisions	Group A; counting rate = 1,000 Nps (min); t = 500 hours	---	---	---	---	---	---
---	Life-test end point	See note 9	---	---	---	---	---	---
(B)	Periodic-check tests							
1031	Variable-frequency vibration	See notes 4 and 11	---	---	---	---	---	---
---	Temperature cycling	See note 8	---	---	---	---	---	---
6205	Leakage current	Ebb = 500 Vdc	---	---	LIb	---	0.5	μAdc
1041	Shock	Hammer angle = 40° (see notes 4 and 13)	---	---	---	---	---	---

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NOTES:

1. The response characteristics, current, and count rate versus field intensity of a typical type 5980 radiation counter tube shall be as shown on figure 3. In determining tube response, Ebb = 700 Vdc and Rp = 1 Meg. Tube current shown as measured by means of a microammeter in series with the tube cathode, and the count rates as determined using a scaler having a resolving time of 0.8 microseconds and a discrimination level of 0.25 volts.

2. A minimum of 28 samples shall be submitted for qualification testing. The samples to be submitted for qualification testing will be broken down into the following subgroups:

(1)	Shelf life only	6
(2)	Electrical, including life test	6
(3)	Shock test only	10
(4)	Variable-frequency vibration	6

3. Each tube shall have an individual serial number permanently marked on the tube.

4. Criterion for passing this test shall be compliance after test of at least 80 percent of the tubes with the requirements for:

   (a) Starting voltage
   (b) Gamma response (count rate)
   (c) Pulse amplitude (1) and (2)
   (d) End-of-plateau voltage
   (e) Relative plateau slope

5. In determining tube response (count rate) Ebb = 700 Vdc and Rp = 1 Meg. Tube count rates shall be determined using a scaler having a resolving time of 5 microseconds and a discrimination level of 0.25 volts.

6. The tube shall be exposed to radiation from a General Electric 15-watt germicidal lamp and a General Electric 15-watt fluorescent lamp, or equivalents, with relative positions as shown on figure 2.

7. Use Government approved standard gamma-test unit.

8. With the tube in a field giving 100 ±10 Nps at 700 Vdc, the tube response (count rate) at 675, 700, and 725 Vdc shall be determined at each of the following temperatures and in the order shown:

   (a) Room temperature
   (b) -40° C
   (c) Room temperature
   (d) 75° C
   (e) Room temperature

   A minimum stabilization time of 30 minutes shall be allowed at each temperature. The absolute count rate at 700 Vdc, and the average relative plateau slope as determined by the readings at the 3 voltages (best average position of straight line), shall not differ from the initial readings at room temperature by more than 10 percent and ±0.1 percent/volt, respectively, at any of the four subsequent temperatures shown.

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NOTES: -Continued

9. The limits for acceptability shall be as follows:

   Background, contamination, and photosensitivity: 12 Npm, maximum
   Starting voltage: Initial value ±10 volts
   Plateau length: 125 volts, minimum.
   Relative plateau slope: 0.25 percent/Vdc, maximum
   End-of-plateau voltage: 750 volts, minimum
   Gamma response (count rate): Initial value ±10 percent
   Pulse amplitude (1) and (2): Initial limits

10. This test to be performed at the conclusion of the holding period.

11. The tube shall be mounted with its longitudinal axis horizontal and perpendicular to the direction of motion by means of a rigid clamp at the center of the cathode. The time for gradually covering the frequency range shall be 15 minutes.

12. Method B (test method 6216); the scaler of note 5 may be utilized for this test.

13. The tube shall be rigidly mounted in positions X and Z by means of a clamp at the center of the cathode. The tube shall be given 10 blows at the X position and 5 at the Z position.

Custodians:
Army - EL
Navy - EC
Air Force - 80

Review activities:
Army -
Air Force - 11, 17,
DSA - ES

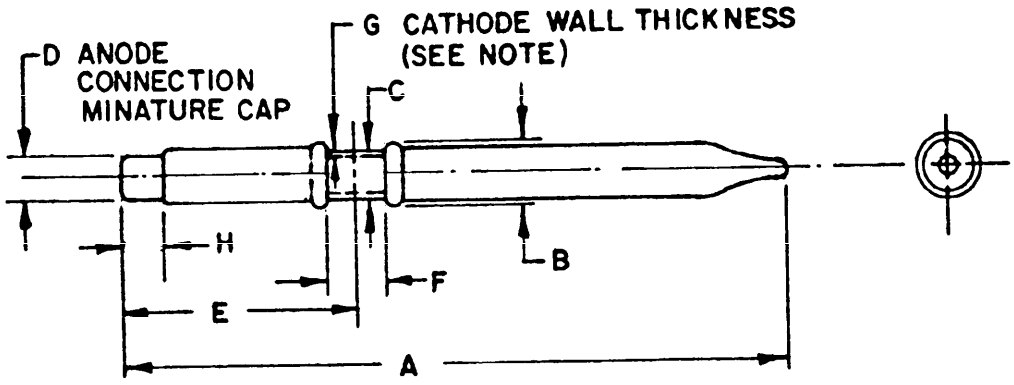
User activities:
Army - ME, WC
Navy - AS, OS, MC, CG, SH

Preparing activity: Navy - EC

Agent: DSA - ES

(Project 5960-2457)

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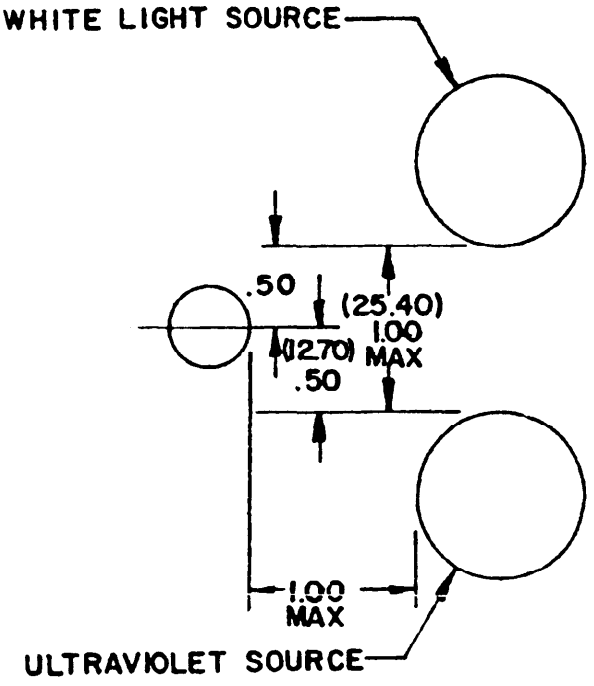


Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
	Quality conformance inspection, part 2	
A	---	4.060(103.12)
B	---	.375(9.53)
E	1.344(34.14)	1.468(37.29)
H	.250(6.35)	---
	Quality conformance inspection, part 3 (periodic-check)	
C	.285(7.24)	.295(7.49)
D	.245(6.22)	.255(6.48)
F	.375(9.53)	---
G	.018(.46)	.022(.56)

NOTE:
Cathode: Allegheny Ludlum Sealmet 1 steel for glass to metal seals, or equivalent.

FIGURE 1. Outline drawing of electron tube type 5980.

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NOTE:
CENTER OF SENSITIVE VOLUME OF TUBE AND CENTERS OF LAMPS LIE IN PLANE OF PAPER

FIGURE 2. Position of tube for photosensitivity test.

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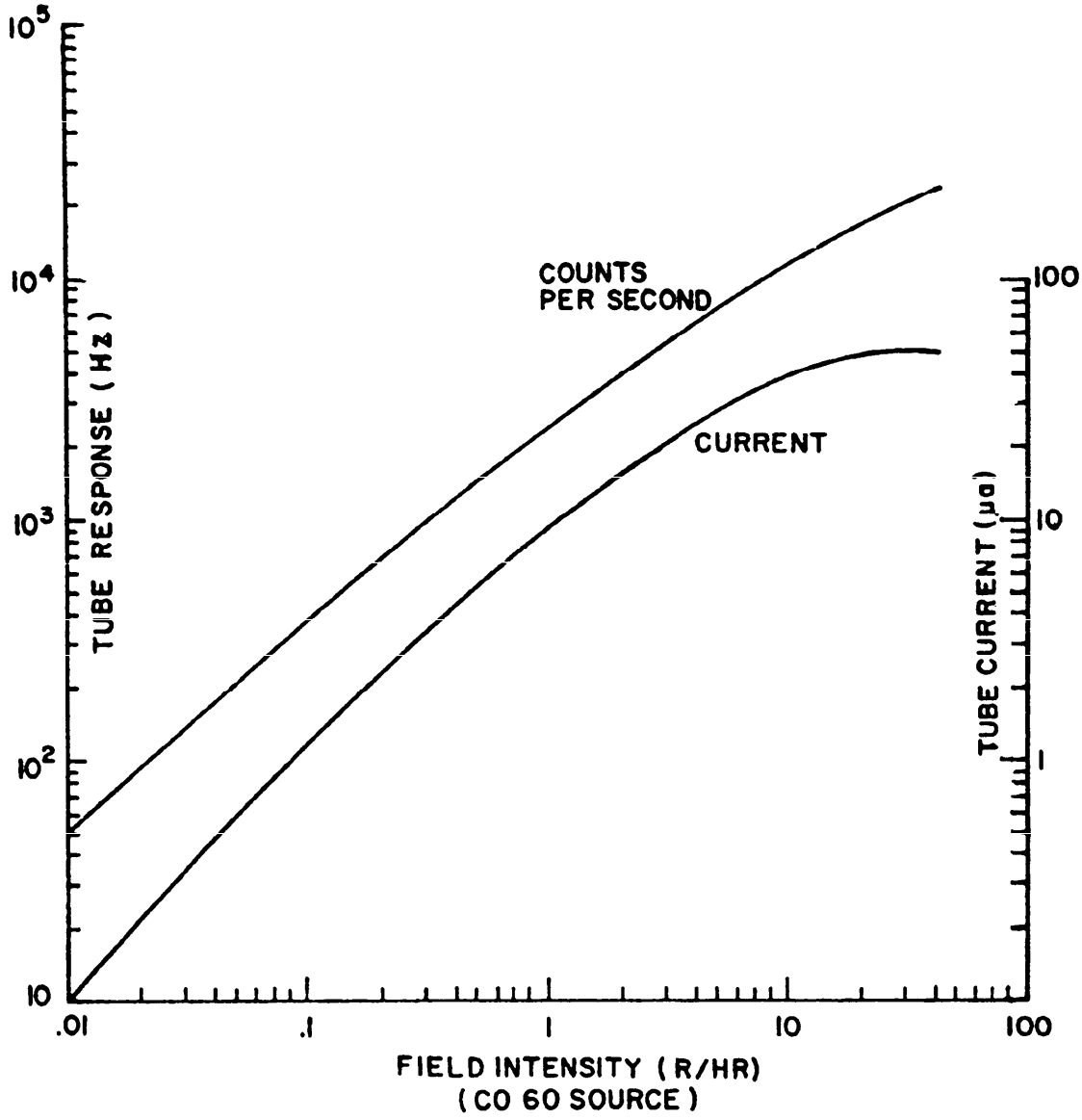


FIGURE 3. Response characteristics of typical type 5980 counter tube.

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